浙江大学实验报告

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课程名称：信号与系统　指导老师：陈晓明　实验类型：Matlab 仿真
实验名称：Lab01　成　绩：　签　名：

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一、 Problem1

1. Question

Problem 1

Consider the signal

$$x_k[n] = \sin(\omega_k n)\,,$$

where $\omega_k = 2\pi k/5$. For $x_k[n]$ given by $k = 1, 2, 4$, and 6, use **stem** to plot each signal on the interval $0 \le n \le 9$. All of the signals should be plotted with separate axes in the same figure using **subplot**. How many unique signals have you plotted? If two signals are identical, explain how different values of ω_k can yield the same signal.

Figure 1: Problem1

2. Code

Listing 1: Problem1 Code

```
%Lab1 P1

clc;clear;

n = 0:9;
k_values = [1, 2, 4, 6];

figure;

for i = 1:length(k_values)
    k = k_values(i);
    wk = 2 * pi * k / 5;
    Tk = sin(wk * n);

```

```
15      subplot(4, 1, i);
16      stem(n, Tk, 'filled');
17  title(['k = ', num2str(k)]);      xlabel('n');
18      ylabel('T_k[n]');
19      grid on;
20    end
21
22    sgtitle('Signals T_k[n] = sin(w_k n) for different k values');
```

3. Figure

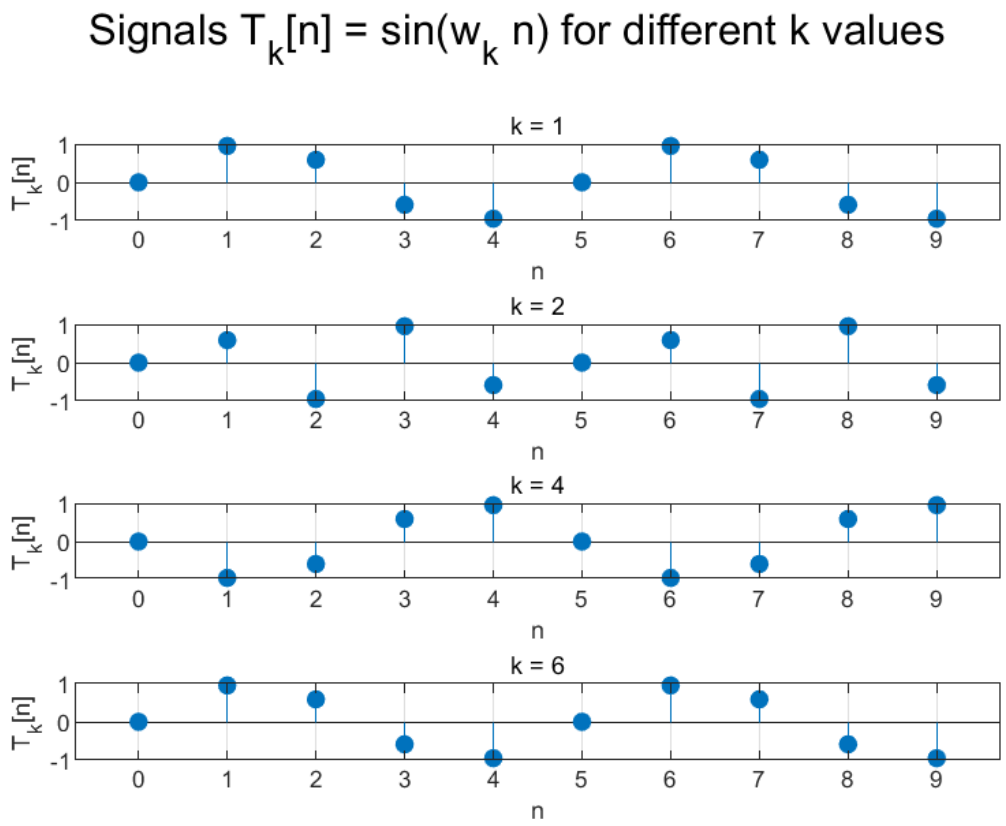


Figure 2: Problem1 Result

4. Answer

There are three unique signals

When different ω values differ by integer multiples of 2π, the corresponding discrete-time signals will be the same.

二、 Problem2

1. Question

Problem 2

Now consider the following three signals

$$x_1[n] = \cos\left(\frac{2\pi n}{N}\right) + 2\cos\left(\frac{3\pi n}{N}\right),$$
$$x_2[n] = 2\cos\left(\frac{2n}{N}\right) + \cos\left(\frac{3n}{N}\right),$$
$$x_3[n] = \cos\left(\frac{2\pi n}{N}\right) + 3\sin\left(\frac{5\pi n}{2N}\right).$$

Assume $N = 6$ for each signal. Determine whether or not each signal is periodic. If a signal is periodic, plot the signal for two periods, starting at $n = 0$. If the signal is not periodic, plot the signal for $0 \le n \le 4N$ and explain why it is not periodic. Remember to use `stem` and to appropriately label your axes.

Figure 3: Problem2

2. Code

Listing 2: Problem2 Code

```
%Lab1 P2

clc;clear;

N = 6;
n_periodic_x1 = 0:24;
n_non_periodic = 0:4*N;
n_periodic_x3 = 0:48;

x1 = cos(2*pi*n_periodic_x1/N) + 2*cos(3*pi*n_periodic_x1/N);
x2 = 2*cos(2*n_non_periodic/N) + cos(3*n_non_periodic/N);
x3 = cos(2*pi*n_periodic_x3/N) + 3*sin(5*pi*n_periodic_x3/2/N);

figure;

subplot(3, 1, 1);
stem(n_periodic_x1, x1, 'filled');
title('x_1[n] = cos(2\pi n/N) + 2cos(3\pi n/N) :period is 2N=12');
xlabel('n');
ylabel('x_1[n]');
grid on;

subplot(3, 1, 2);
```

```
  stem(n_non_periodic, x2, 'filled');
title('x_2[n] = 2cos(2n/N)+ cos(3n/N):Not periodic');  xlabel('n');
  ylabel('x_2[n]');
  grid on;

  subplot(3, 1, 3);
  stem(n_periodic_x3, x3, 'filled');
  title('x_3[n] = cos(2\pi n/N) + 3sin(5\pi n/2N) :period is 4N=24');
  xlabel('n');
  ylabel('x_3[n]');
  grid on;

  sgtitle('Periodicity Analysis of Signals');
```

3. Figure

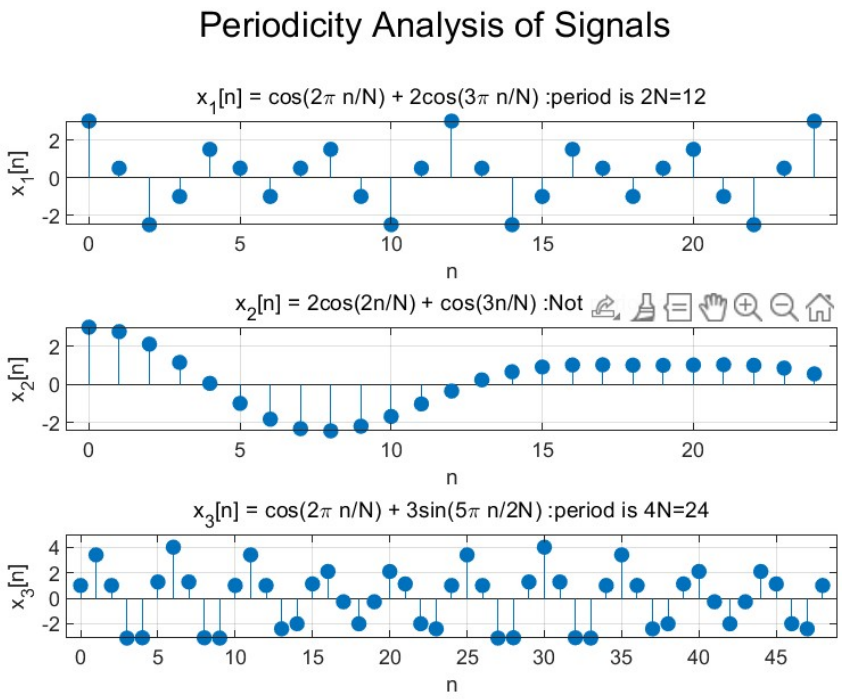


Figure 4: Problem2 Result

4. Answer

$x_1[n]$: $T = 2N = 12$
$x_2[n]$: $T1 = N\pi$ and $T2 = \frac{2N}{3}\pi$;$T = 2N * \pi$,which is irrational,so $x_2[n]$ isn't periodic
$x_3[n]$: $T = 4N = 24$

三、 Problem3

1. Question

Problem 3

Plot each of the following signals on the interval $0 \leq n \leq 31$:

$$x_1[n] = \sin\left(\frac{\pi n}{4}\right) \cos\left(\frac{\pi n}{4}\right),$$
$$x_2[n] = \cos^2\left(\frac{\pi n}{4}\right),$$
$$x_3[n] = \sin\left(\frac{\pi n}{4}\right) \cos\left(\frac{\pi n}{8}\right).$$

What is the fundamental period of each signal? For each of these three signals, how could you have determined the fundamental period without relying upon MATLAB?

Figure 5: Problem3

2. Code

Listing 3: Problem3 Code

```
%Lab1 P3

clc;clear;

n = 0:31;

x1 = sin(pi*n/4) .* cos(pi*n/4);
x2 = cos(pi*n/4).^2;
x3 = sin(pi*n/4) .* cos(pi*n/8);

figure;

subplot(3,1,1);
stem(n, x1,'filled');
title('Signal x1[n] = sin(\pi n / 4) cos(\pi n / 4)');
xlabel('n');
ylabel('x1[n]');

subplot(3,1,2);
stem(n, x2,"filled");
title('Signal x2[n] = cos^2(\pi n / 4)');
xlabel('n');
```

```
  ylabel('x2[n]');
subplot(3,1,3);  stem(n, x3,"filled");
  title('Signal x3[n] = sin(\pi n / 4) cos(\pi n / 8)');
  xlabel('n');
  ylabel('x3[n]');
```

3. Figure

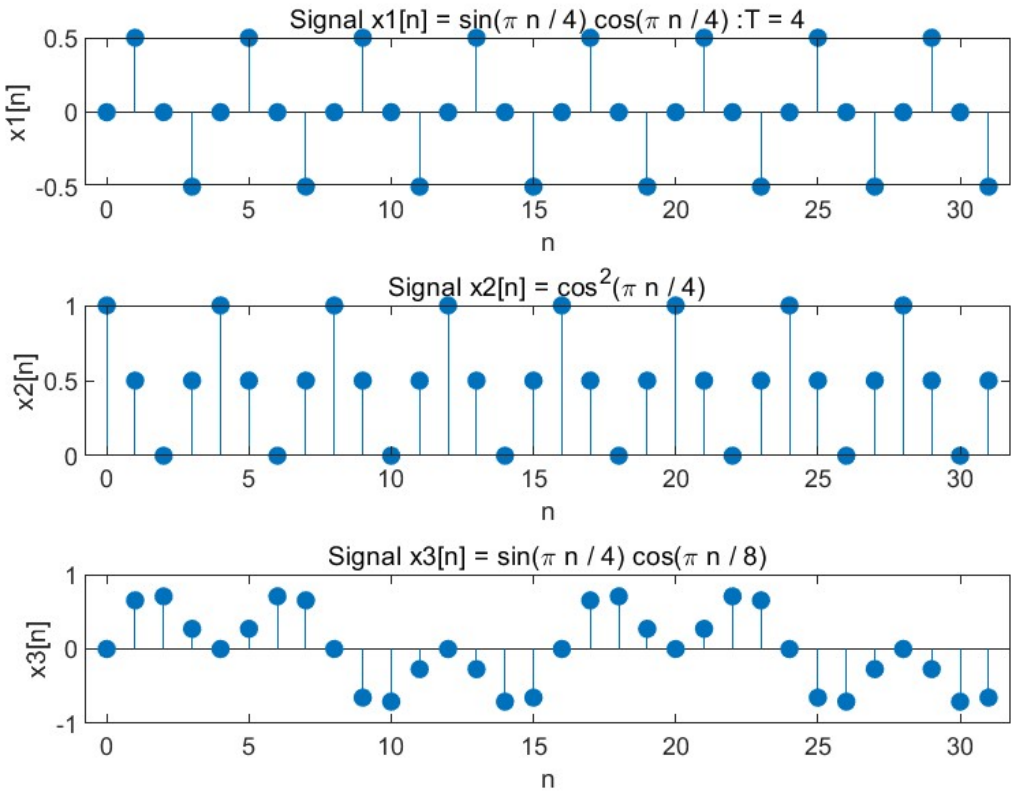


Figure 6: Problem3 Result

4. Answer

$$x_1 = \frac{1}{2}sin(\frac{\pi n}{2}) \Rightarrow x_1[n] : T = 4$$

$$x_2 = \frac{1}{2}[1 + cos(\frac{\pi n}{2})] \Rightarrow x_2[n] : T = 4$$

$$x_3 = \frac{1}{2}[sin(\frac{3\pi n}{8}) + sin(\frac{\pi n}{8})] \Rightarrow x_3[n] : T = 16$$

四、 Problem4

1. Question

Problem 4

Consider the signals you plotted in Problem 2 and 3. Is the addition of two periodic signals necessarily periodic? Is the multiplication of two periodic signals necessarily periodic? Clearly explain your answers.

Figure 7: Problem4

2. Answer

(1) The addtion of two periodic signals isn't necessarily periodic

for continuous signal:

$$f(t) = cos2\pi t + cost$$

$$f(t+T) = cos(2\pi t + 2\pi T) + cos(t+T)$$

$$T = k \ and \ T = 2k\pi, k \in N$$

it is contradictory, so isn't necessarily periodic
for discrete signal,it is periodic

(2) The multiplication of two periodic signals isn't necessarily periodic

for continuous signal:

$$f(t) = cos\frac{2\pi+1}{2}tcos\frac{2\pi-1}{2}t = \frac{1}{2}(cos2\pi t + cost)$$

which is not periodic, so isn't necessarily periodic
for discrete signal,it is periodic

五、 Problem5

1. Question

Problem 5

Basic Problems

For these problems, you are told which property a given system does not satisfy, and the input sequence or sequences that demonstrate clearly how the system violates the property. For each system, define MATLAB vectors representing the input(s) and output(s). Then, make plots of these signals, and construct a well reasoned argument explaining how these figures demonstrate that the system fails to satisfy the property in question.

(a). The system $y[n] = \sin((\pi/2)x[n])$ is not linear. Use the signals $x_1[n] = \delta[n]$ and $x_2[n] = 2\delta[n]$ to demonstrate how the system violates linearity.

(b). The system $y[n] = x[n] + x[n+1]$ is not causal. Use the signal $x[n] = u[n]$ to demonstrate this. Define the MATLAB vectors x and y to represent the input on the interval $-5 \leq n \leq 9$, and the output on the interval $-6 \leq n \leq 9$, respectively.

Figure 8: Problem5

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2. Code

Listing 4: Problem5a Code

```
%Lab1 P5a

clc;clear;
n = -5:5;
delta = (n == 0);
x1 = delta;
x2 = 2*delta;
y1 = sin(pi/2 * x1);
y2 = sin(pi/2 * x2);

figure;

subplot(2,1,1);
stem(n, y1,"filled");
title('y1[n] = sin(\pi/2 * x1[n])');
xlabel('n');
ylabel('y1[n]');

subplot(2,1,2);
stem(n, y2,"filled");
```

```
  title('y2[n] = sin(\pi/2 * x2[n])');
xlabel('n');  ylabel('y2[n]');
  sgtitle('Analysis of Signals Property :y_2[0]!=2y_1[0],not linear');
```

Listing 5: Problem5b Code

```
  %Lab1 P5 b

  clc;clear;
  n = -5:9;
  u = @(n) double(n >= 0);
  x = u(n);
  y = u(n)+u(n+1);

  figure;
  subplot(2,1,1);
  stem(n, x,"filled");
  title('Input x[n] = u[n]');
  xlabel('n');
  ylabel('x[n]');

  subplot(2,1,2);
  stem(n, y,"filled");
  title('Output y[n] = x[n] + x[n+1]');
  xlabel('n');
  ylabel('y[n]');
```

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3. Figure

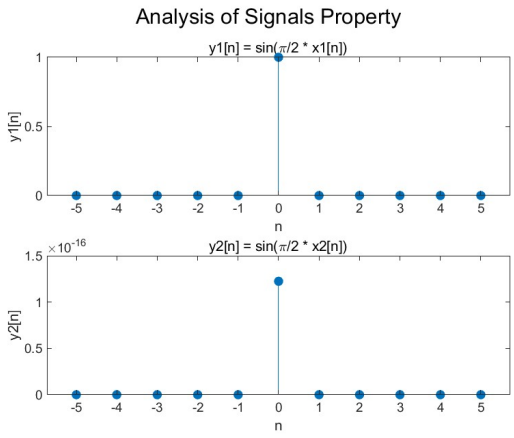


Figure 9: Problem5a Result

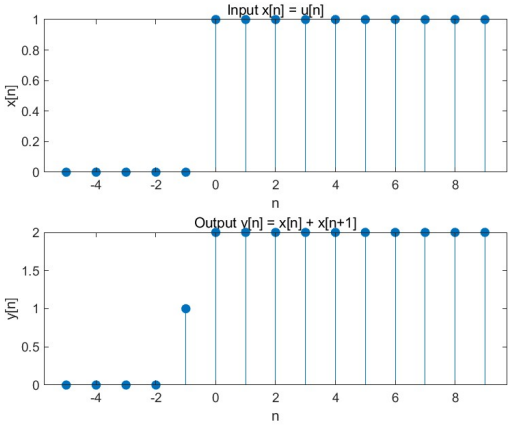


Figure 10: Problem5b Result

4. Answer

P5a:Non-Linear Verification:

$$y_2[n] \neq 2y_1[n],$$

P5b:Non-Causality Verification: The system uses x[n+1] (future input) to compute y[n]

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六、 Problem6

1. Question

Problem 6

(a). Write a function `y=diffeqn(a,x,yn1)` which computes the output $y[n]$ of the causal system determined by Eq. (1.6). The input vector `x` contains $x[n]$ for $0 \leq n \leq N-1$ and `yn1` supplies the value of $y[-1]$. The output vector `y` contains $y[n]$ for $0 \leq n \leq N-1$. The first line of your M-file should read

```
function y = diffeqn(a,x,yn1)
```

Hint: Note that $y[-1]$ is necessary for computing $y[0]$, which is the first step of the autoregression. Use a `for` loop in your M-file to compute $y[n]$ for successively larger values of n, starting with $n = 0$.

(b). Assume that $a = 1$, $y[-1] = 0$, and that we are only interested in the output over the interval $0 \leq n \leq 30$. Use your function to compute the response due to $x_1[n] = \delta[n]$ and $x_2[n] = u[n]$, the unit impulse and unit step, respectively. Plot each response using `stem`.

Figure 11: Problem6

figure 装 订

2. Code

Listing 6: Problem6 Code

```
%Lab1 P6

function y = diffeqn(a, x, yn1)
    N = length(x);
    y = zeros(1, N);
    y(1) = a * yn1 + x(1);

    for i = 2:N
        y(i) = a * y(i-1) + x(i);
    end
end

n = 0:30;
x1 = (n == 0);
x2 = (n >= 0);
yn1 = 0;
a = 1;
y1 = diffeqn(a, x1, yn1);
```

```
  y2 = diffeqn(a, x2, yn1);
figure;  subplot(2,1,1);
  stem(n, y1,"filled");
  title('Response to x1[n] = \delta[n]');
  xlabel('n');
  ylabel('y1[n]');

  subplot(2,1,2);
  stem(n, y2,"filled");
  title('Response to x2[n] = u[n]');
  xlabel('n');
  ylabel('y2[n]');
```

3. Figure

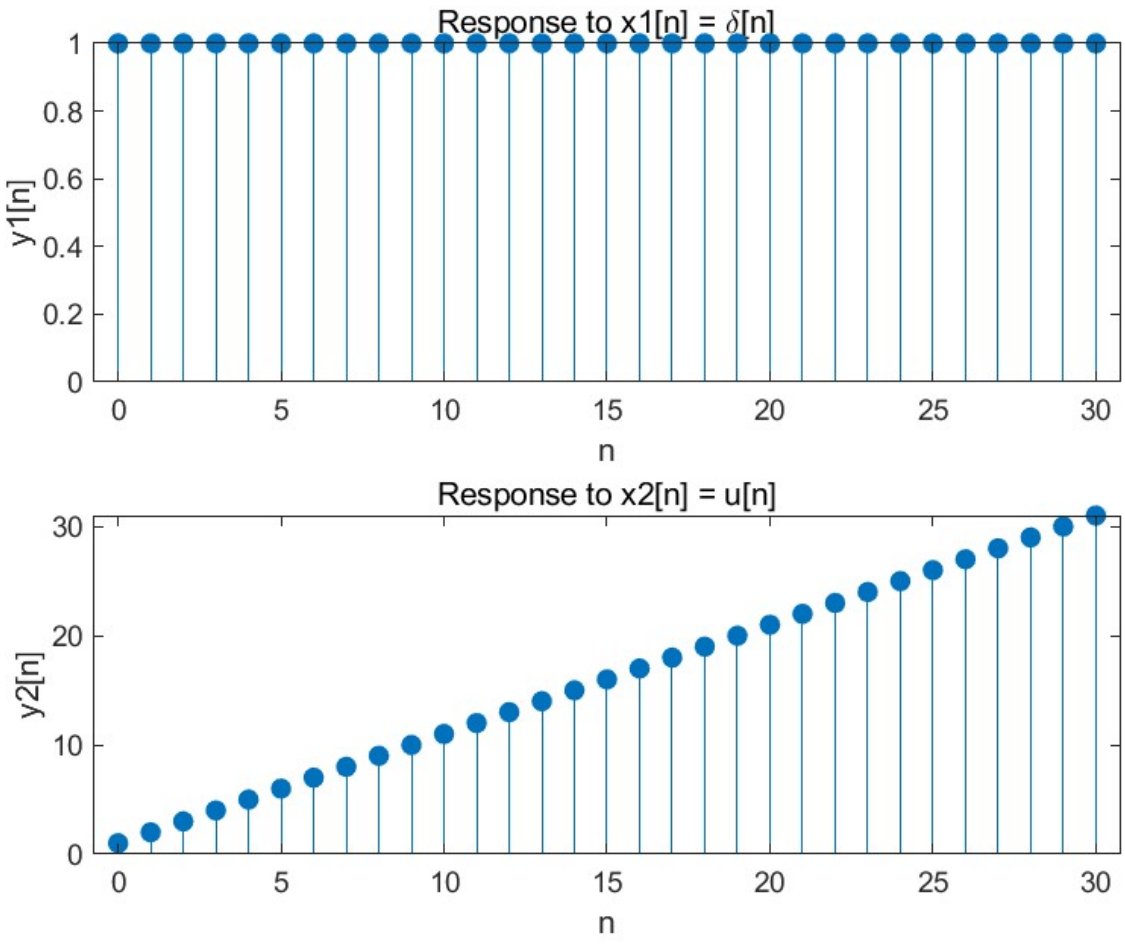


Figure 12: Problem6 Result

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七、 Problem7

1. Question

Problem 7

Consider the continuous-time sinusoid

$$x(t) = \sin(2\pi t/T)\,.$$

A symbolic expression can be created to represent $x(t)$ within MATLAB by executing

```
>>  x = sym('sin(2*pi*t/T)');
```

The variables of **x** are the single character strings **'t'** and **'T'**. The function **ezplot** can be used to plot a symbolic expression which has only one variable, so you must set the fundamental period of $x(t)$ to a particular value. If you desire $T = 5$, you can use **subs** as follows

```
>>  x5 = subs(x,5,'T');
```

Thus **x5** is a symbolic expression for $\sin(2\pi t/5)$. Create the symbolic expression for **x5** and use **ezplot** to plot two periods of $\sin(2\pi t/5)$, beginning at $t = 0$. If done correctly, your plot should be as shown in Figure 1.2.

Figure 13: Problem7

2. Code

Listing 7: Problem7 Code

```
%Lab1 P7

syms t T
x = sin(2*pi*t/T);

x5 = subs(x, T, 5);

ezplot(x5, [0, 10])
grid on
xlabel('t')
ylabel('x(t)')
title('Plot of sin(2\pi t / 5) over two periods')
```

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3. Figure

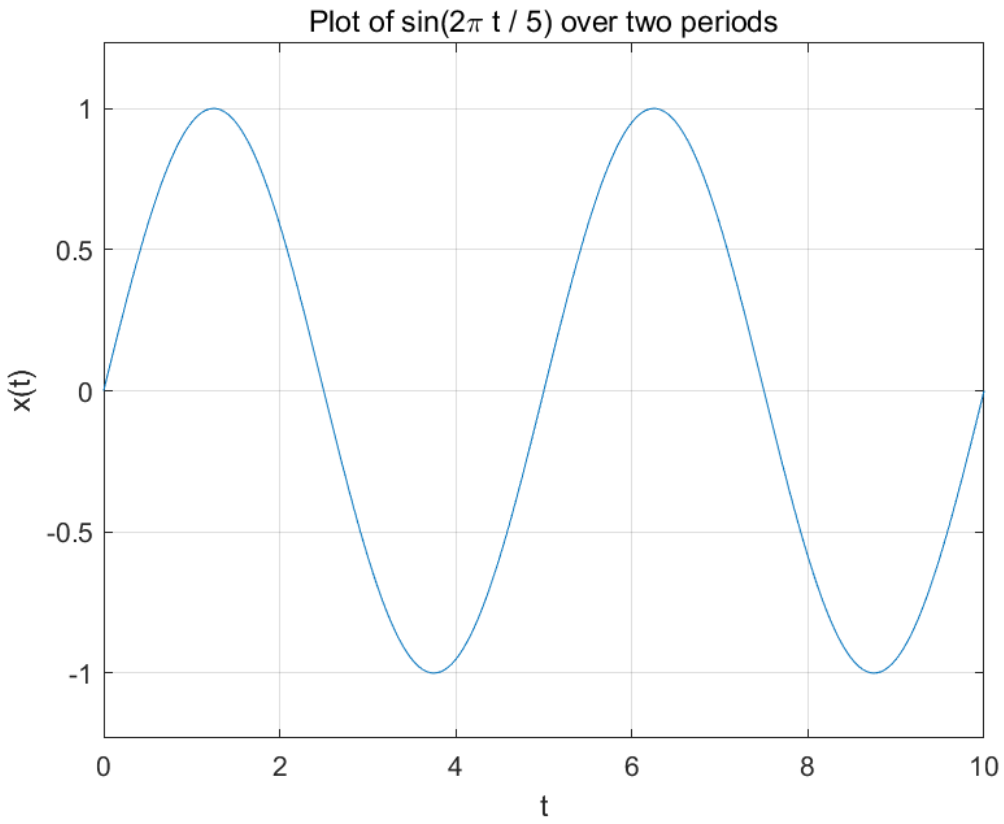


Figure 14: Problem7 Result

装 订 xr

八、 Problem8

1. Question

Problem 8

In the following problems you will write M-files for extracting the real and imaginary components, or the magnitude and phase, of a symbolic expression for a complex signal.

(d). Store in `x` a symbolic expression for the signal

$$x(t) = e^{i2\pi t/16} + e^{i2\pi t/8}.$$

Remember to use `'i'` rather than `'j'` within symbolic expressions to represent $\sqrt{-1}$. The function `ezplot` cannot be used directly for plotting $x(t)$, since $x(t)$ is a complex signal. Instead, the real and imaginary components must be extracted and then plotted separately.

(e). Write a function `xr=sreal(x)` which returns a symbolic expression `xr` representing the real part of $x(t)$. If your function is working properly, `ezplot(xr)` will plot the real component of $x(t)$. Similarly, write a function `xi=simag(x)` which returns a symbolic expression `xi` representing the imaginary component of $x(t)$. The first line of the M-file `sreal.m` should be

```
function xr = sreal(x)
```

You can then use `compose('real(x)',x)` to create a symbolic expression for the real component of $x(t)$. Use `ezplot` and the functions you created to plot the real and imaginary components of $x(t)$ on the interval $0 \le t \le 32$. Use a separate plot for each component. What is the fundamental period of $x(t)$?

Figure 15: Problem8

2. Code

Listing 8: Problem8 Code

```
%Lab1 P8

function xr = sreal(x)
xr = real(x);
end

function xi = simag(x)
  xi = imag(x);
end

syms t
x = exp(1i*2*pi*t/16) + exp(1i*2*pi*t/8);
```

```
13
14 xr = sreal(x);  xi = simag(x);
15
16   figure;
17   ezplot(xr, [0, 32])
18   grid on
19   xlabel('t')
20   ylabel('Re\{x(t)\}')
21   title('Real Component of x(t)')
22
23   figure;
24   ezplot(xi, [0, 32])
25   grid on
26   xlabel('t')
27   ylabel('Im\{x(t)\}')
28   title('Imaginary Component of x(t)')
```

3. Figure

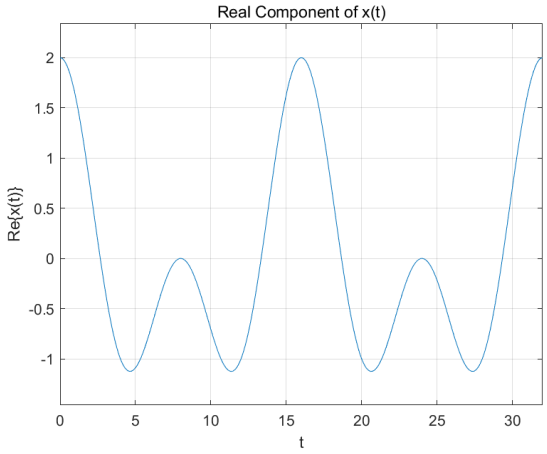


Figure 16: Problem8a Result

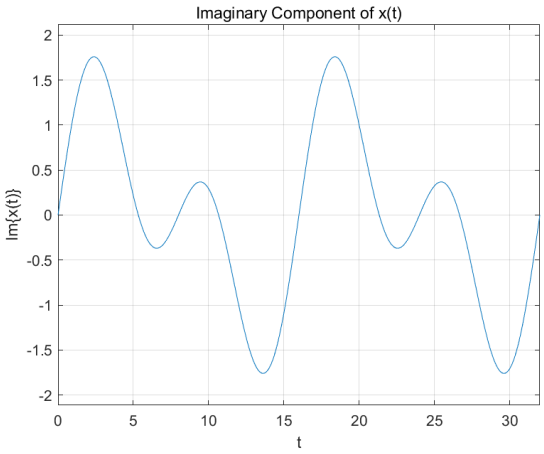


Figure 17: Problem8b Result

4. Answer

The fundamental period of $x(t)$ *is* 16